

FIITJEE – Phase Test-4 (JEE-**PHYSICS, CHEMISTRY & MATHEMATICS****CPT1-1****Q.P. CODE:0000****PAPER - 1****13/08/2023****Time Allotted: 3 Hours****Maximum Marks: 183**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains **Three Sections**.
3. **Section-I** is Physics, **Section-II** is Chemistry and **Section-III** is Mathematics.
4. Each **Section** is further divided into **Two Parts: Part-A & C** in the OMR. Part-B of OMR to be left unused.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with **Blue/Black Ball Point Pen** for each character of your Enrolment No. and write in ink your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Two Parts.

- (i) **Part-A (01-07)** – Contains seven (07) multiple choice questions which have **One or More** correct answer.
Full Marks: +4 If only the bubble(s) corresponding to all the correct option(s) is (are) darkened.
Partial Marks: +1 For darkening a bubble corresponding to **each correct option**, provided NO incorrect option is darkened.
Zero Marks: 0 If none of the bubbles is darkened. **Negative Marks: –1** In all other cases.
 For example, if (A), (C) and (D) are all the correct options for a question, darkening all these three will result in **+4 marks**; darkening only (A) and (D) will result in **+2 marks**; and darkening (A) and (B) will result in **–1 marks**, as a wrong option is also darkened.
- (ii) **Part-A (08-13)** – Contains six (06) multiple choice questions which have **ONLY ONE CORRECT** answer. Each question carries **+3 marks** for correct answer and **-1 marks** for wrong answer.
- (iii) **Part-C (01-05)** contains five (05) Numerical based questions with single digit integer as answer, ranging from 0 to 9 (both inclusive) and each question carries **+3 marks** for correct answer. **There is no negative marking.**

Name of the Candidate : _____

Batch : _____ **Date of Examination :** _____

Enrolment Number : _____

Useful Data Chemistry:

Gas Constant R = $8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
 = $0.0821 \text{ Lit atm K}^{-1} \text{ mol}^{-1}$
 = $1.987 \approx 2 \text{ Cal K}^{-1} \text{ mol}^{-1}$

Avogadro's Number N_a = 6.023×10^{23}

Planck's Constant h = $6.626 \times 10^{-34} \text{ Js}$
 = $6.25 \times 10^{-27} \text{ erg.s}$

1 Faraday = 96500 Coulomb

1 calorie = 4.2 Joule

1 amu = $1.66 \times 10^{-27} \text{ kg}$

1 eV = $1.6 \times 10^{-19} \text{ J}$

Atomic No : H=1, D=1, Li=3, Na=11, K=19, Rb=37, Cs=55, F=9, Ca=20, He=2, O=8, Au=79.

Atomic Masses: He=4, Mg=24, C=12, O=16, N=14, P=31, Br=80, Cu=63.5, Fe=56, Mn=55, Si = 28 Pb=207, u=197, Ag=108, F=19, H=2, Cl=35.5

Useful Data Physics:

Acceleration due to gravity $g = 10 \text{ m/s}^2$

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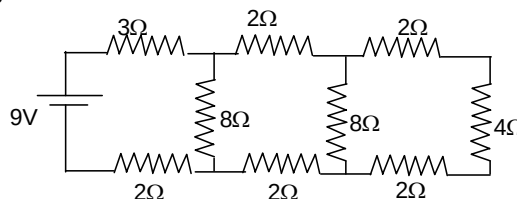
Section – I (Physics)

PART – A

One or More Than One Correct Type

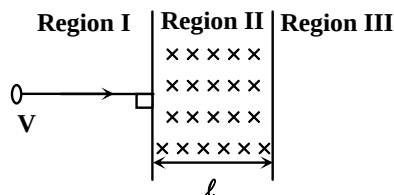
1. A positively charged particle of specific charge α (charge per unit mass) is released at time $t = 0$ from the origin with an initial velocity of $\vec{v} = v_0 \hat{i}$ in a uniform magnetic field of $\vec{B} = -B_0 \hat{k}$. Then
- (A) x-component of velocity of the particle at time t is $v_0 \cos B_0 \alpha t$.
 (B) y-component of velocity of the particle at time t is $v_0 \sin B_0 \alpha t$.
 (C) the path of the particle is a circle of radius $\frac{v_0}{B_0 \alpha}$.
 (D) the path of the particle is a helix of uniform pitch.

2. In the circuit shown in figure below the current through
- (A) the 3Ω resistor is 1 A
 (B) the 3Ω resistor is 0.2 A
 (C) the 4Ω resistor is 0.50 A
 (D) the 4Ω resistor is 0.25 A



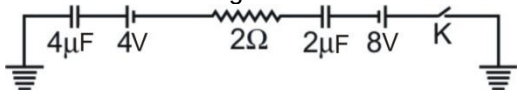
3. A wire is bent to form a square loop of side a . It carries a current I and is kept perpendicular to a uniform magnetic field. Such that magnetic moment is in the same direction as the field. Shape of the loop is changed from square to a circle without changing the length of the wire and current. Then:
- (A) the decrease in magnetic moment of the square loop with respect to circular loop is $ia^2(1 - 4/\pi)$.
 (B) the decrease in magnetic moment of the square loop which respect to circular loop is $ia^2(1 - 2/\pi)$.
 (C) the amount of work done in doing so is $iBa^2(\pi - 4)/\pi$.
 (D) the amount of work done in doing so is $ia^2(\pi - 4)/4$.

4. A particle of mass m and charge q , moving with velocity V enters Region II normal to the boundary as shown in the figure. Region II has a uniform magnetic field B perpendicular to the plane of the paper. The length of the Region II is ℓ . Choose the correct choice(s).



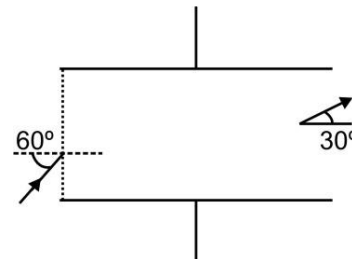
- (A) The particle enters Region III only if its velocity $V > \frac{q\ell B}{m}$
 (B) The particle enters Region III only if its velocity $V < \frac{q\ell B}{m}$
 (C) Path length of the particle in Region II is maximum when velocity $V = \frac{q\ell B}{m}$
 (D) Time spent in Region II is same for any velocity V as long as the particle returns to Region I

5. As shown in the figure find the correct statement/s :



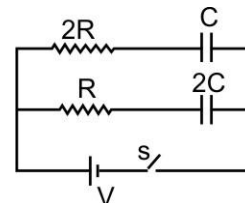
- (A) i at instant switch is closed is 2A
 (B) i after long time is 2A
 (C) charge on $4\mu F$ after long time is $\frac{16}{3}\mu C$ (when switch is closed)
 (D) charge on $2\mu F$ at instant switch is closed is $8\mu C$

6. An alpha particle enters in the region between the plates of capacitor at an angle 60° and leaves the region at an angle of 30° , then



- (A) The ratio of speeds entering and leaving the region is $\sqrt{3}:1$
 (B) The ratio of K.E. entering and leaving the region is 1 : 3
 (C) The path inside the region is circular
 (D) The path inside the region is parabolic

7. In the shown circuit, switch s is closed at time $t = 0$. Select the correct statement



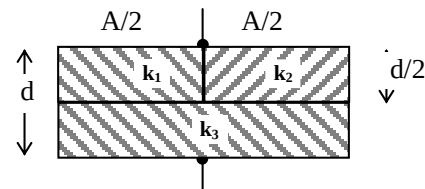
- (A) Rate of increase of charge is same in both the capacitors
 (B) Ratio of charge stored in capacitors C and 2C at any time t is 1:2
 (C) time constant of both the capacitors are equal
 (D) Steady state charge on capacitors C and 2C are in ratio 1:2

Single Correct Choice Type

8. An insulating thin rod of length ℓ has a linear charge density $\rho(x) = \rho_0 \frac{x}{\ell}$ on it. The rod is rotated about an axis passing through the origin ($x = 0$) and perpendicular to the rod. If the rod makes n rotations per second, then the time averaged magnetic moment of the rod is:

- (A) $\pi n \rho_0 \ell^3$ (B) $\frac{\pi}{3} n \rho_0 \ell^3$ (C) $\frac{\pi}{4} n \rho_0 \ell^3$ (D) $n \rho_0 \ell^3$

9. A parallel plate capacitor of area A , plate separation d and capacitance C is filled with three different dielectric materials having dielectric constants k_1 , k_2 and k_3 as shown. If a single dielectric material is to be used to have the same capacitance C in this capacitor, then its dielectric constant k is given by



- (A) $\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{2k_3}$ (B) $\frac{1}{k} = \frac{1}{k_1 + k_2} + \frac{1}{2k_3}$
 (C) $k = \frac{k_1 k_2}{k_1 + k_2} + 2k_3$ (D) $k = k_1 + k_2 + 2k_3$

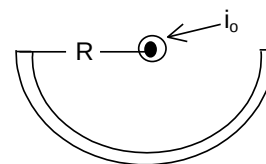
10. Charge is distributed within a sphere of radius R with a volume charge density $\rho(r) = \frac{A}{r^2} e^{-2r/a}$, where A and a are constants. If Q is the total charge of this charge distribution, the radius R is:

- (A) $a \log \left(1 - \frac{Q}{2\pi a A} \right)$ (B) $\frac{a}{2} \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right)$

$$(C) a \log \left(\frac{1}{1 - \frac{Q}{2\pi a A}} \right)$$

$$(D) \frac{a}{2} \log \left(1 - \frac{1}{2\pi a A} \right)$$

11. Shown in the figure is a very long semicylindrical conducting shell of radius R and carrying a current i . An infinitely long straight current carrying conductor is lying along the axis of the semicylinder. If the current flowing through the straight wire be i_0 , then the force per unit length on the conducting wire is :



$$(A) \frac{\mu_0 i i_0}{\pi R^2}$$

$$(B) \frac{\mu_0 i_0 i}{\pi R^2}$$

$$(C) \frac{\mu_0 i i_0}{\pi^2 R}$$

(D) None of these

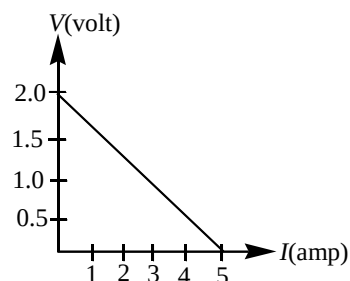
12. For a cell, a graph is plotted between the potential difference V across the terminals of the cell and the current I drawn from the cell. The e.m.f and the internal resistance of the cell are E and r , respectively. Then

$$(A) E = 2V, r = 0.5\Omega$$

$$(B) E = 2V, r = 0.4\Omega$$

$$(C) E > 2V, r = 0.5\Omega$$

$$(D) E > 2V, r = 0.4\Omega$$



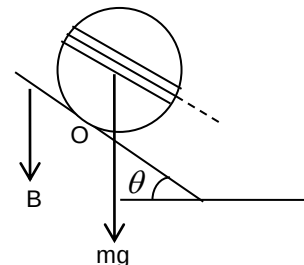
13. In figure, a coil of single turn is wound on a sphere of radius r and mass m . The plane of the coil is parallel to the inclined plane and lies in the equatorial plane of the sphere. If the sphere is in rotational equilibrium, the value of B is: [current in the coil is I]

$$(A) \frac{mg}{\pi Ir}$$

$$(B) \frac{mg \sin \theta}{\pi Ir}$$

$$(C) \frac{mg \cos \theta}{\pi Ir}$$

(D) None of these



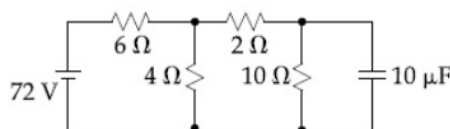
PART – C Integer Type

1. A parallel plate capacitor of capacitance C has spacing d between two plates having area A . The region between the plates is filled with N dielectric layers, parallel to its plates, each with thickness $\delta = \frac{d}{N}$. The dielectric constant of the m^{th} layer is $K_m = K \left(1 + \frac{m}{N} \right)$. For a very large

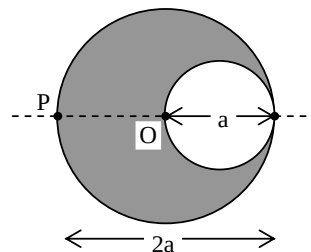
$N (> 10^3)$, the capacitance C is $\alpha \left(\frac{K \epsilon_0 A}{d \ln 2} \right)$. The value of α will be _____.

[ϵ_0 is the permittivity of free space]

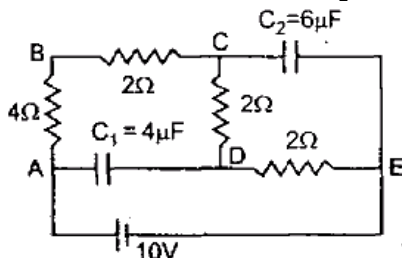
2. The charge on the capacitor in the following circuit is $100k \mu C$. Find the value of k .



3. A cylindrical cavity of diameter a exists inside a cylinder of diameter $2a$ as shown in the figure. Both the cylinder and the cavity are infinity long. A uniform current density J flows along the length. If the magnitude of the magnetic field at the point P is given by $\frac{N}{12} \mu_0 a J$, then the value of N is



4. Charge $Q = 2 \times 10^{-3} \text{ C}$ is uniformly distributed over the curved surface of cone of semi vertical angle 45° and height 10 cm. Cone is rotated uniformly about its axis at angular velocity of 10 r/s. Then its magnetic dipole moment is $n \times 10^{-5}$, Find n .
5. Potential difference across C_2 in steady state is _____ V.



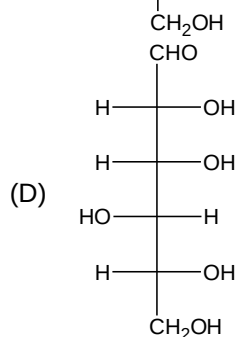
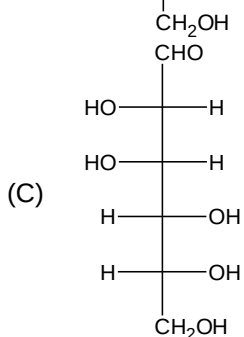
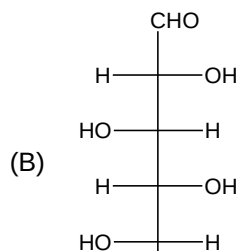
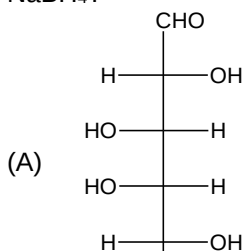
space for rough work

Section – II (Chemistry)

PART – A

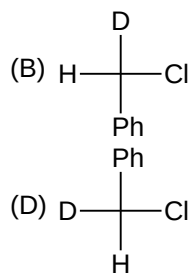
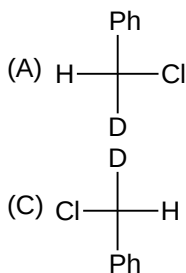
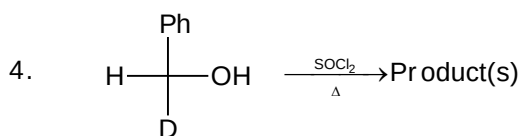
One or More Than One Correct Type

1. Which of the following gives an optically active compound when reacted with sodium borohydride, NaBH_4 ?



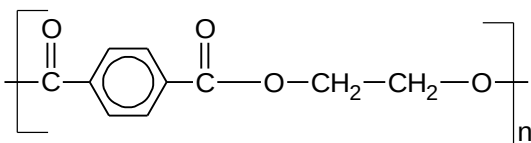
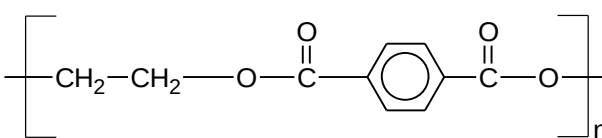
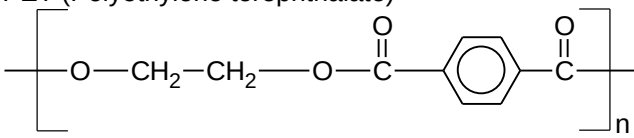
2. Which of the following contain only addition homopolymers?
 (A) Neoprene (B) Dacron (C) Bakelite (D) PVC

3. $\text{Alkene} + \text{H}_2\text{O} \xrightarrow{\text{H}^+} \text{Alcohol}$
 Which of the following statements about this reaction is correct?
 (A) It is acid-catalysed addition of water to alkene.
 (B) It occurs through carbonium ion intermediate.
 (C) There can be rearrangement in the mechanism.
 (D) Butanol is the major product when the alkene is butene.



5. Which of the following can reduce Fehling's solution
 (A) Formic acid (B) 1-hydroxy-3-pentanone
 (C) 1-methoxy ethanol (D) Acetaldehyde

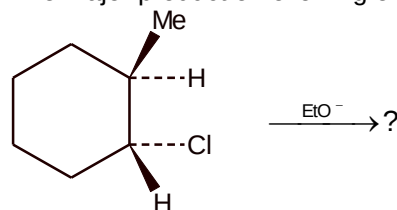
6. A carboxylic acid X is treated with bromine (1mole) in the presence of a little phosphorus. The isolated product is heated with a strong base and then oxidized with $O_3 / Zn-H_2O$. One of the two oxidation products is a neutral compound which reacts with hydroxylamine to give only single oxime Y. What is/are possible structure of X can be
- (A) $CH_3CH_2CH(CH_3)CH_2CO_2H$ (B) $CH_3CH_2CH(CH_3)CO_2H$
 (C) $CH_3CH_2CO_2H$ (D) $(CH_3)_2CHCH_2CO_2H$
7. Which of the following statement(s) is/are incorrect

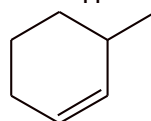
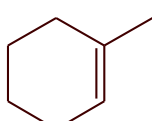
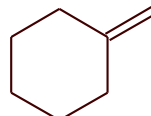
- (A)  is not the structure of terylene
- (B)  is not the structure of polymer
 PET (Polyethylene terephthalate)
- (C)  is incorrect structure of Dacron
- (D) Melamine is trimer of cyanamide ($N \equiv C - NH_2$)

PART – A: (Single Correct Answer Type)

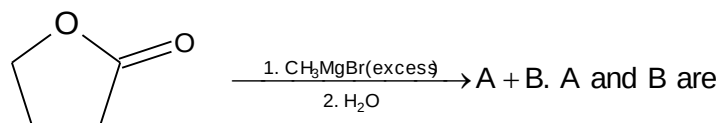
This section contains **06 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

8. The major product of following elimination reaction will be



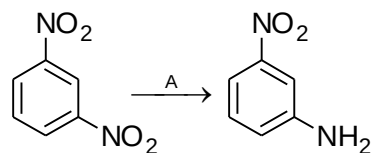
- (A)  (B) 
 (C)  (D) Reaction not possible

9.



- (A) $\text{H}_3\text{C}-\text{OH}$,
- (B) ,
- (C) ,
- (D) ,

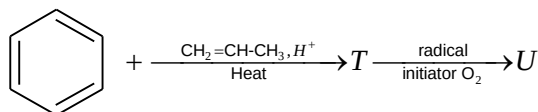
10.



A is

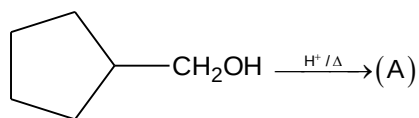
- (A) Sn/HCl (B) LiAlH_4 (C) NH_4HS (D) $\text{Zn} + \text{HCl}$

11. The major product U in the following reactions is:



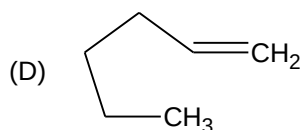
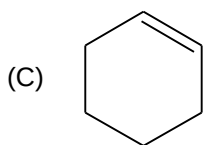
- (A)
- (B)
- (C)
- (D)

12.

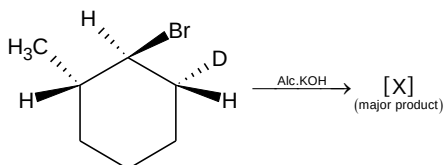


Compound (A) is

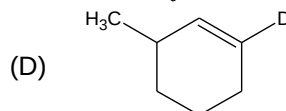
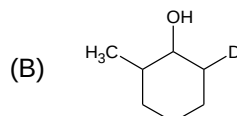
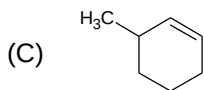
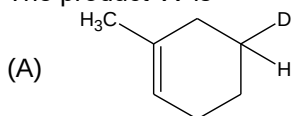
- (A)
- (B)



13.

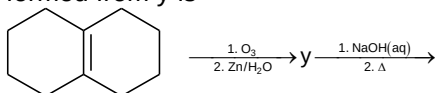


The product 'X' is



PART - C Integer Type

- 0.1 mol of a hydroxyl compound reacts with 62.5 g of PCl_5 (mol. wt. 208.5). Determine the number of $-\text{OH}$ groups.
- Number of enantiomeric pairs obtained on monochlorination of 2-methyl butane
- In the scheme given below, the total number of intramolecular aldol condensation products formed from y is



- Number of aldol product between the reaction acetaldehyde and propanaldehyde is (exclude stereo isomers)
- How many aldehydes and ketones are possible with formula $\text{C}_5\text{H}_{10}\text{O}$? (exclude stereo isomers)

space for rough work

Section – III (Mathematics)

PART – A

One or More Than One Correct Type

1. In which of the following function Lagrange's mean value theorem is applicable in $[0, 1]$
- (A) $f(x) = \begin{cases} x - \frac{1}{2}, & x < \frac{1}{2} \\ \left(x - \frac{1}{2}\right)^2, & x \geq \frac{1}{2} \end{cases}$ (B) $g(x) = \begin{cases} \frac{\sin x}{x}, & x \neq 0 \\ 1, & x = 0 \end{cases}$
- (C) $h(x) = x|x|$ (D) $p(x) = |x|$
2. A function $g(x)$ is defined as $g(x) = \frac{1}{4}f(2x^2 - 1) + \frac{1}{2}f(1 - x^2)$ and $f'(x)$ is an increasing function then $g(x)$ is increasing in the interval
- (A) $\left(-\sqrt{\frac{2}{3}}, 0\right)$ (B) $\left(\sqrt{\frac{2}{3}}, \infty\right)$ (C) $(1, 10)$ (D) none of these
3. Let $\int \frac{dx}{x^{2008} + x} = \frac{1}{p} \ln\left(\frac{x^q}{1 + x^r}\right) + c$ where $p, q, r \in \mathbb{N}$ then
- (A) $q = r$ (B) $q + r = 2p$ (C) $\frac{p}{q} = \frac{2q}{r}$ (D) $p + q + r = 6021$
4. If $f(x) = \lim_{n \rightarrow \infty} \frac{x}{x^{2n} + 1}$, then..
- (A) $f(1^+) + f(1^-) = 0$ (B) $f(1^+) + f(1^-) + f(1) = 3/2$
- (C) $f(-1^+) + f(-1^-) = -1$ (D) $f(1^+) + f(-1^-) = 0$
5. Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $g: \mathbb{R} \rightarrow \mathbb{R}$ and $h: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable functions such that $f(x) = x^3 + 3x + 2$, $g(f(x)) = x$ and $h(g(g(x))) = x$ for all $x \in \mathbb{R}$. Then
- (A) $g'(2) = \frac{1}{3}$ (B) $h'(1) = 666$ (C) $h(0) = 16$ (D) $h(g(3)) = 38$
6. The value of $\int_0^\pi \frac{\sin\left(n + \frac{1}{2}\right)x}{\sin\left(\frac{x}{2}\right)} dx$, $n \in \mathbb{N}$, is
- (A) independent of n (B) $2n\pi$
- (C) π (D) $\frac{n\pi}{2}$
7. Let $F(x) = \int_0^x (t-1)(t-2)^2 dt$, then
- (A) $\left(1, -\frac{17}{12}\right)$ is a point of minimum (B) at $x = 2$ concavity of $F(x)$ will changed
- (C) $f(x)$ increases in $(1, \infty)$ (D) $(1, -3)$ is a point of minimum

Single Correct Choice Type

8. If $f(x) = \frac{1}{2}x - 1$, then in the interval $[0, \pi]$. (where $[\cdot]$ is G.I.F.)
 (A) $\tan [f(x)]$ and $\frac{1}{f(x)}$ are both continuous (B) $\tan [f(x)]$ and $\frac{1}{f(x)}$ are both discontinuous
 (C) $\tan [f(x)]$ and $f^{-1}(x)$ are both continuous (D) $\tan [f(x)]$ is continuous but $\frac{1}{f(x)}$ is discontinuous.
9. For the curve $y = be^{x/a}$
 (A) Length of sub-tangent is constant (B) Length of sub-normal is constant
 (C) Length of tangent is constant (D) Length of normal is constant
10. If $\int (\sin 3\theta + \sin \theta) e^{\sin \theta} \cos \theta d\theta = (A \sin^3 \theta + B \cos^2 \theta + C \sin \theta + D \cos \theta + E) e^{\sin \theta} + F$, then
 (A) $B = 12$ (B) $D = 1$ (C) $B = -10$ (D) $A = -4$
11. If $y = f(x)$ and $y = g(x)$ are symmetrical about the line $x = \frac{\alpha + \beta}{2}$, then $\int_{\alpha}^{\beta} f(x) g'(x) dx$ is equal to
 (A) $\int_{\alpha}^{\beta} f'(x) g(x) dx$ (B) $3 \int_{\alpha}^{\beta} f'(x) g(x) dx$
 (C) $\frac{1}{2} \int_{\alpha}^{\beta} (f(x) g'(x) - f'(x) g(x)) dx$ (D) $\frac{1}{2} \int_{\alpha}^{\beta} (f(x) g'(x) + f'(x) g(x)) dx$
12. Let $f(x) = \int_0^x e^{t-[t]} dt$ ($x > 0$), where $[x]$ denotes greatest integer function less than or equal to x , is
 (A) continuous and differentiable $\forall x \in (0, 3]$ (B) continuous but not differentiable $\forall x \in (0, 3]$
 (C) $f(1) = e$ (D) $f(2) = 2(e + 1)$
13. The function f is continuous and has the property $f(f(x)) = 1 - x \forall x \in [0, 1]$ and $J = \int_0^1 f(x) dx$ then which of the following is incorrect?
 (A) $f\left(\frac{1}{4}\right) + f\left(\frac{3}{4}\right) = 1$ (B) the value of $J = \frac{1}{2}$
 (C) $f\left(\frac{1}{3}\right) + f\left(\frac{2}{3}\right) = 2$ (D) $\frac{2}{\pi} \int_0^{\pi/2} \frac{\sin x dx}{(\sin x + \cos x)} = J$

PART – C
Integer Type

1. Let f be a one - one function with domain $\{1, 2, 3\}$ and range $\{x, y, z\}$. It is given that exactly one of the following statements is true and the remaining two are false. $f(1) = x$, $f(2) \neq x$, $f(3) \neq y$. Then $f^{-1}(x)$ is....
2. If $\int \frac{(2x+3)dx}{x(x+1)(x+2)(x+3)+1} = \frac{-1}{ax^2+bx+c} + k$, then $2a + b + c$ is...

3. If $\int_{-1}^4 f(x) dx = 4$ and $\int_2^4 (3 - f(x)) dx = 7$, then the value of $\int_{-1}^2 f(x) dx$ is...
4. The point on the curve $y = 2x^2 - 5x + 3$ between the points A (1,0) and B(2,1) where the tangent is parallel to the chord AB is (α, β) then $(6\alpha + 7\beta)$ is.....
5. Let $f(n)$ be a continuous function in $\mathbb{R}$ such that $f\left(\frac{1}{4n}\right) = (\sin e^n) e^{-n^2} + \frac{n^2}{n^2 + 1}$. Then the value of $f(0)$ is...

space for rough work

FIITJEE COMMON TEST

BATCHES: CTY 2224 X LOT
CLASS- XII
PHASE TEST – 4 (13-08-2023)
QP. Code: 00000
JEE ADVANCED PAPER-1
ANSWER KEY

SECTION - I: PHYSICS

PART – A

- | | | | |
|------------|---------|---------|------------|
| 1. A, B, C | 2. A, D | 3. A, C | 4. A, C, D |
| 5. A, C | 6. A, D | 7. BCD | 8. C |
| 9. B | 10. B | 11. C | 12. B |
| 13. A | | | |

PART – C

- | | | | |
|------|------|------|------|
| 1. 1 | 2. 2 | 3. 5 | 4. 5 |
| 5. 4 | | | |

SECTION - II: CHEMISTRY

PART – A

- | | | | |
|------------|---------|------------|---------|
| 1. B, C, D | 2. A, D | 3. A, B, C | 4. A, C |
| 5. A, C, D | 6. CD | 7. A, B, C | 8. A |
| 9. D | 10. C | 11. B | 12. C |
| 13. C | | | |

PART – C

- | | | | |
|------|------|------|------|
| 1. 3 | 2. 2 | 3. 1 | 4. 4 |
| 5. 7 | | | |

SECTION- III: MATHEMATICS

PART – A

- | | | | |
|---------------|------------|------------|------------|
| 1. B, C, D | 2. A, B, C | 3. A, C, D | 4. B, C, D |
| 5. A, B, C, D | 6. A, C | 7. A, B, C | 8. B |
| 9. A | 10. D | 11. C | 12. B |
| 13. C | | | |

PART – C

- | | | | |
|------|------|------|------|
| 1. 2 | 2. 6 | 3. 5 | 4. 9 |
| 5. 1 | | | |

HINTS & SOLUTIONS**SECTION - I: PHYSICS****PART – A**

1. A, B, C

2. A, D

3. A, C

4. A, C, D

Sol. $r = \frac{mv}{qB}$, particle enters region III if $r > \ell$

$$\text{or } \frac{mv}{qB} > \ell, v > \frac{qB\ell}{m}$$

Path length is πr , maximum when r is maximum

$$\text{or } r = \ell \Rightarrow v = \frac{qB\ell}{m}$$

$$\text{Time period } t = \frac{\pi r}{v} = \frac{\pi m}{qB} = \text{constant}$$

5. A, C

Sol. $-\frac{q}{4} - 4 - \frac{q}{2} + 8 = 0$

$$\Rightarrow q = \frac{16}{3}$$

$$\text{at } t = \infty \quad i = 0$$

$$\text{at } t = 0 \Rightarrow -4 - 2i + 8 = 0 \Rightarrow i = 2A$$

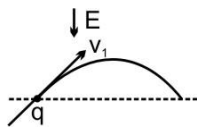
6. A, D

Sol.



$$v_1 \cos 60^\circ = v_2 \cos 30^\circ$$

$$\frac{v_1}{v_2} = \frac{\cos 30^\circ}{\cos 60^\circ} = \frac{\sqrt{3}}{2 \times \frac{1}{2}} = \sqrt{3}$$



7. BCD

Sol. Charge on C,

$$q_1 = CV \left(1 - e^{-\frac{t}{2RC}} \right)$$

And on 2C

$$q_2 = 2CV \left(1 - e^{-\frac{t}{2RC}} \right)$$

8. C

Sol. $dM = di A$

$$= \left(\frac{dq\omega}{2\pi} \right) \pi x^2$$

$$= (\rho dx) \frac{\omega}{2\pi} \pi x^2$$

$$M = \int_0^L dM$$

9. B

Sol. $C_3 = (\epsilon_0 A k_3) / (d/2)$; then, $C = (C_1 + C_2)C_3 / (C_1 + C_2 + C_3)$
equating 'C' to $(k\epsilon_0 A/d)$, we get $1/k = \{1/(k_1 + k_2)\} + 1/2k_3$

10. B

Sol. $Q = \int \rho 4\pi r^2 dr = \int_0^R \left(\frac{A}{r^2} e^{-\frac{2r}{a}} \right) (4\pi r^2) dr$

$$= 4\pi A \frac{a}{2} \left(1 - e^{-\frac{2R}{a}} \right)$$

$$\Rightarrow R = \frac{-a}{2} \log \left(1 - \frac{Q}{2\pi A a} \right)$$

11. C

Sol. The net magnetic force on the conducting wire per unit length

$$= F = \int 2dF \cos \theta$$

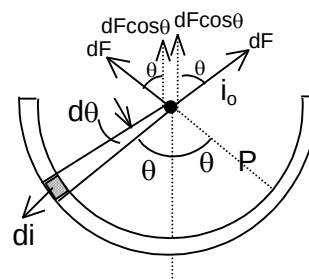
$$\Rightarrow F = \int 2 \left[\frac{\mu_0 (di) i_o}{2\pi R} \right] \cos \theta$$

$$\Rightarrow F = \frac{\mu_0 i_o}{\pi R} \int di \cos \theta$$

$$\text{when } di = \frac{i}{\pi R} \times R d\theta = \frac{id\theta}{\pi}$$

$$\Rightarrow F = \frac{\mu_0 i_o}{\pi R} \int \frac{(id\theta) \cos \theta}{\pi}$$

$$\Rightarrow F = \frac{\mu_0 i_o i}{\pi^2 R} \int_0^{\pi/2} \cos \theta d\theta = \frac{\mu_0 i_o i}{\pi^2 R}$$



12. B

Sol. $V = E - ir$. When $i = 0$, the potential reading is 2V. Hence e.m.f = 2V
When $V = 0, i = 5A$. This gives $r = 0.4\Omega$

13. A

PART - C

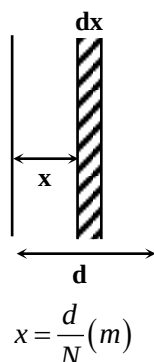
1. 1

Sol. $\frac{1}{dC} = \frac{dx}{k \left(1 + \frac{m}{N} \right) \epsilon_0 A}$

$$\int \frac{1}{dC} = \int_0^d \frac{dx}{k \epsilon_0 \left(1 + \frac{x}{d} \right)}$$

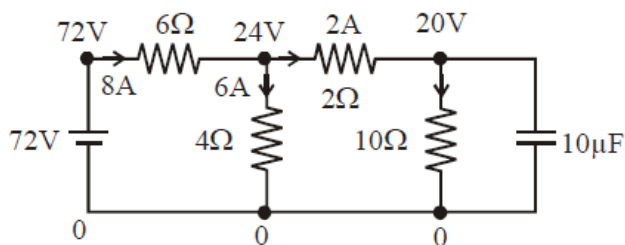
$$\Rightarrow C_{eq} = \frac{K \epsilon_0 A}{d \ln 2}$$

$$\alpha = 1.00$$



2. 2

- Sol. Different potential is shown at different points.
 $q = eV$
 $q = 10\mu F \times 20 = 200 \mu C$



3. 5

Sol.
$$B = \frac{\mu_0(J\pi a^2)}{2\pi a} - \frac{\mu_0(J\pi a^2/4)}{2\pi\left(\frac{3a}{2}\right)}$$

$$B = \frac{5\mu_0 Ja}{12} = \frac{\mu_0 NJa}{12}$$

So $N = 5$

4. 5

Sol. Using relation $\vec{\mu} = \frac{q\vec{L}}{2m}$

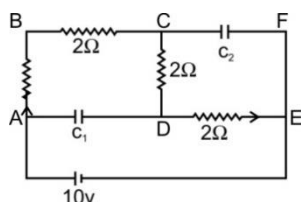
Where L is angular momentum of conical shell

We get $\vec{\mu} = \frac{q\omega}{4} h^2 \tan^2 \theta$

$$\mu = \frac{2 \times 10^{-3} \times 10}{4} \left(\frac{10}{100}\right)^2 \times 1 = 5 \times 10^{-5}$$

5. 4

Sol.



P.D. across C_2 is $V_C - V_F = V_C - V_E$

$$V_C - V_E = (V_C - V_D) + (V_D - V_E)$$

By K.L.L.

$$10 = i[4 + 2 + 2 + 2]$$

$$i = 1A$$

$$\therefore V_C - V_E = 4V$$

SECTION - II: CHEMISTRY

PART - A

- 1.

SECTION- III: MATHEMATICS

PART - A

1. B, C, D

Sol. $f(x)$ is not differentiable at $x = \frac{1}{2}$

2. A, B, C

Sol. $g'(x) = x(f'(2x^2 - 1) - f'(1 - x^2))$

$$\text{If } x > 0, f'(2x^2 - 1) - f'(1 - x^2) > 0 \Rightarrow 2x^2 - 1 > 1 - x^2$$

$$\text{So, } x \in \left(\sqrt{\frac{2}{3}}, \infty \right) \text{ and If } x < 0, f'(1 - x^2) > f'(2x^2 - 1) \Rightarrow 1 - x^2 > 2x^2 - 1$$

$$\text{So } x \in \left(-\sqrt{\frac{2}{3}}, 0 \right)$$

3. A, C, D

$$\text{Sol. } \int \frac{dx}{x^{2008} + x} = \frac{1}{2007} \ln \left(\frac{x^{2007}}{1 + x^{2007}} \right) + c$$

4. B, C, D

5. A, B, C, D

$$\text{Sol. } g'(2) = \frac{1}{f'(0)} = \frac{1}{3}$$

$$\begin{aligned} h(g(g(x))) &= x & \text{let } g(g(x)) &= y \\ h(y) &= f(f(y)) & g(x) &= f(y) \\ h'(y) &= f'(f(x)) \cdot f'(y) & x &= f(f(y)) \\ h(0) &= 16, & h'(1) &= 666 \end{aligned}$$

6. A, C

$$\text{Sol. Let } I_n = \int_0^\pi \frac{\sin\left(n + \frac{1}{2}\right)x}{\sin\left(\frac{x}{2}\right)} dx \Rightarrow I_{n+1} - I_n = \int_0^\pi \frac{\sin\left(n + \frac{1}{2}\right)x - \sin\left(n + \frac{3}{2}\right)x}{\sin\left(\frac{x}{2}\right)} dx$$

$$= -2 \int_0^\pi \cos(n+1)x dx = -\frac{2}{n+1} [\sin(n+1)x]_0^\pi = 0$$

$$\Rightarrow I_0 = I_1 = I_2 = I_3 = \dots = I_n$$

$$\text{hence } I_n = I_0 = \int_0^\pi dx = \pi.$$

7. A, B, C

$$\text{Sol. } F'(x) = (x-1)(x-2)^2, \text{ so } F'(x) = 0 \text{ implies that } x = 1 \text{ and } x = 2$$

Around $x = 1$, $F'(x)$ changes sign from negative to positive.

$$\text{Hence } F(x) \text{ is minimum at } x = 1 \text{ and } F(1) = \int_0^1 (t^3 - 5t^2 + 8t - 4) dt$$

$$= \frac{1}{4} - \frac{5}{3} + 4 - 4 = -\frac{17}{12}. \text{ Hence } \left(1, -\frac{17}{12}\right) \text{ is a point of minimum.}$$

$$\text{Also, } F''(x) = (x-2)(3x-4) \Rightarrow x = 2, \frac{4}{3} \text{ are points of inflection.}$$

8. B

$$\text{Sol. } [f(x)] = \left[\frac{x}{2} - 1 \right] = -1, \quad 0 \leq x < 2$$

$$= 0, \quad 2 \leq x \leq \pi$$

$$\tan[f(x)] = \begin{cases} \tan(-1) = -\tan 1, & 0 \leq x < 2 \\ \tan 0 = 0, & 2 \leq x \leq \pi \end{cases}$$

so $\tan[f(x)]$ is discontinuous at $x = 2$.

The function $\frac{1}{f(x)} = \frac{1}{\frac{x}{2} - 1}$ is discontinuous at $x = 2$

$f^{-1}(x) = 2x + 1$,
so $f^{-1}(x)$ is continuous on $[0, \pi]$

9. A

Sol. Curve $y = be^{x/a}$

$$\frac{dy}{dx} = \frac{b}{a} e^{x/a} \Rightarrow y' = \frac{y}{a}$$

Length of subtangent $= \frac{y_1}{y'} = \frac{y}{y/a} = a$ which is constant.

10. D

Sol. $\int e^t \cdot (4t - 4t^3) dt = (At^3 - Bt^2 + Ct + B + E) e^t + F$

$D = 0$ as $f(t)$ is polynomial in t

$A = -4, B = -12$

11. C

Sol. $f(x) = f(\alpha + \beta - x)$

$\Rightarrow f(\alpha) = f(\beta)$ and $g(\alpha) = g(\beta)$

$$\int_{\alpha}^{\beta} f(x)g'(x)dx = f(x)g(x)\Big|_{\alpha}^{\beta} - \int_{\alpha}^{\beta} f'(x)g(x)dx$$

$$2I = \int_{\alpha}^{\beta} f(x)g'(x)dx - \int_{\alpha}^{\beta} f'(x)g(x)dx$$

$$I = \frac{1}{2} \int_{\alpha}^{\beta} (f(x)g'(x) - f'(x)g(x))dx$$

12. B

Sol. We have $f(x) = \int_0^x e^{t-t} dt = \int_0^x e^{t^2} dt$, so

$$f(x) = \begin{cases} \int_0^x e^t dt & \text{if } x \in (0,1) \\ \int_0^1 e^t dt + \int_1^x e^{t-1} dt & \text{if } x \in [1,2) \\ \int_0^1 e^t dt + \int_1^2 e^{t-1} dt + \int_2^x e^{t-2} dt & \text{if } x \in [2,3) \end{cases} \Rightarrow f(x) = \begin{cases} e^x - 1 & \text{if } x \in (0,1) \\ (e-1) + (e^{x-1} - 1) & \text{if } x \in [1,2) \\ 2(e-1) + (e^{x-2} - 1) & \text{if } x \in [2,3) \end{cases}$$

Clearly $f(x)$ is continuous $\forall x > 0$ but not differentiable $\forall x \in \mathbb{N}$

Also $f(2) = 2(e-1) = 0 = 2(e-1)$

13. C

Sol. Given $f(f(x)) = -x + 1$

Replacing $x \rightarrow f(x)$

$$f(f(f(x))) = -f(x) + 1$$

$$f(1-x) = -f(x) + 1$$

$$f(x) + f(1-x) = 1 \dots\dots (1) \Rightarrow \textbf{(A)}$$

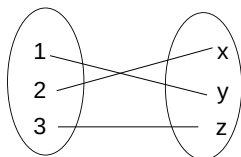
$$\text{Now } J = \int_0^1 f(x)dx = \int_0^1 f(1-x)dx \text{ (Using King)}$$

$$2J = \int_0^1 (f(x) + f(1-x))dx; 2J = \int_0^1 dx = 1 \Rightarrow J = \frac{1}{2} \text{ Ans.}$$

PART – C

1. 2

Sol.



2. 6

Sol. $\int \frac{(2x+3)dx}{x(x+1)(x+2)(x+3)+1} = \frac{-1}{x^2+3x+1} + k$

3. 5

Sol. $\int_{-1}^4 f(x)dx = 4, \int_2^4 (3-f(x))dx = 7$

$$\Rightarrow 6 - \int_2^4 f(x)dx = 7.$$

$$\int_{-1}^2 f(x)dx = \int_{-1}^4 f(x)dx - \int_2^4 f(x)dx = 4 - (-1) = 5$$

Hence, $\int_{-1}^2 f(x)dx = 5$

4. 9

Sol. Using LMVT, $f'(c) = \frac{f(b)-f(a)}{b-a}$ so $4c-5 = \frac{1-0}{2-1}$ $c=3/2$ so point is $\left(\frac{3}{2}, 0\right)$

5. 1

Sol. As f is continuous so $f(0) = \lim_{n \rightarrow \infty} f\left(\frac{1}{4n}\right)$

$$f(0) = \lim_{n \rightarrow \infty} f(1/4n) = \lim_{n \rightarrow \infty} \left((\sin e^n) e^{-n^2} + \frac{1}{1+1/n^2} \right) = 0 + 1 = 1$$